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SCHOOL OF ENGINEERING

GROUP OF **(ROBOTICS & AI ENGINEERING: SEC 1)**



LEAF COLLECTING ROBOT

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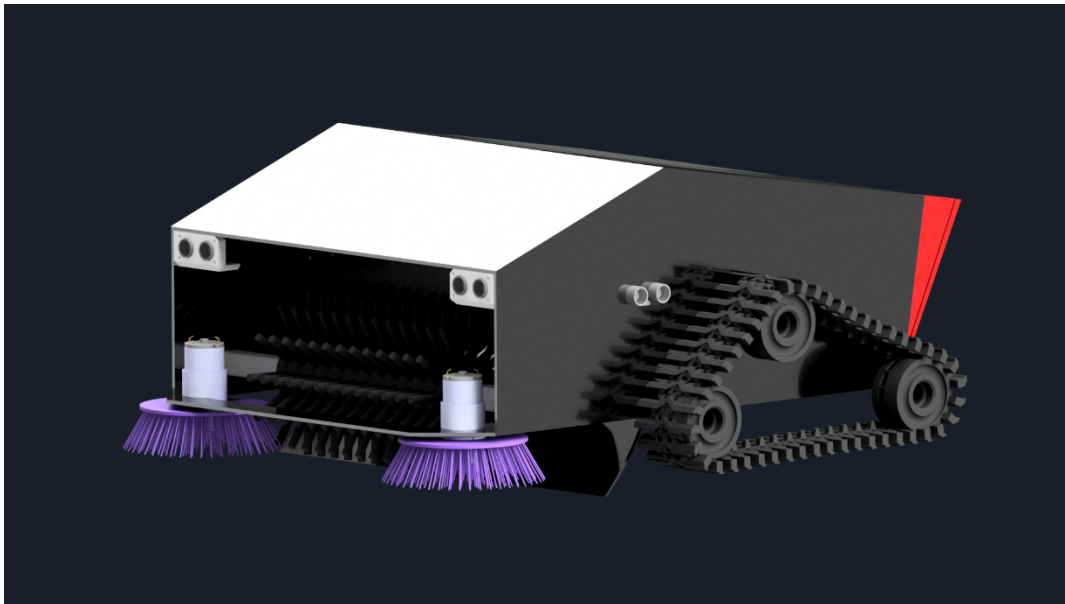
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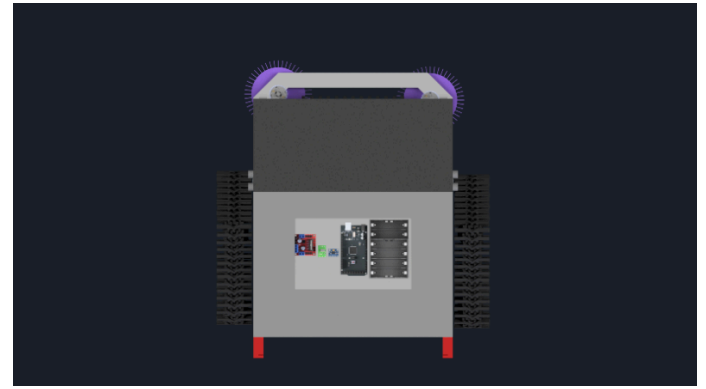
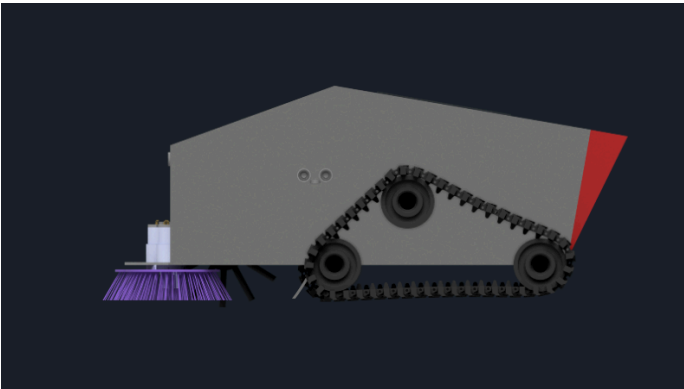
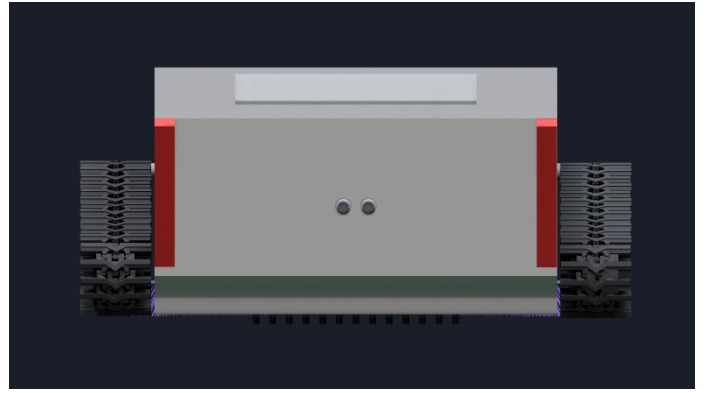
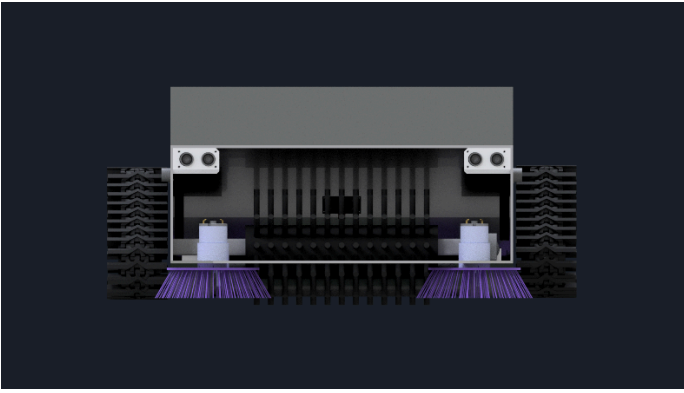
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1. CONCEPTS AND DRAWINGS

1.1 Concept design

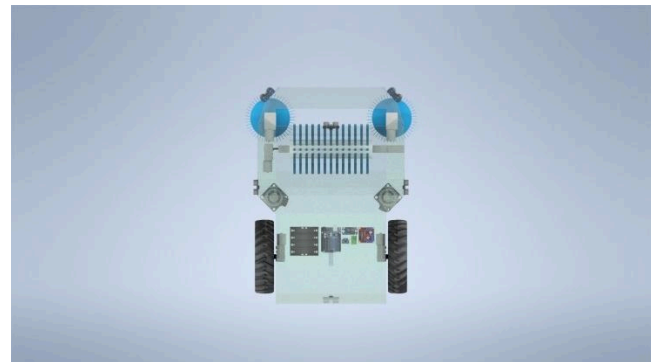
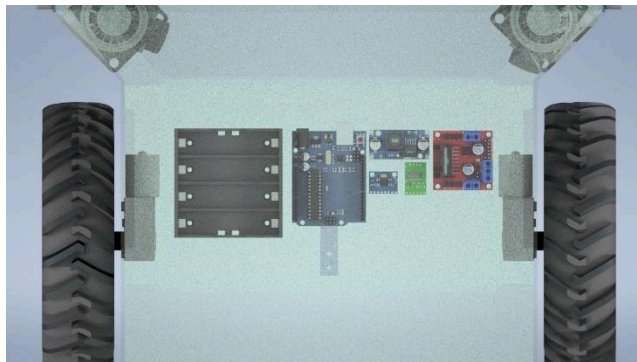
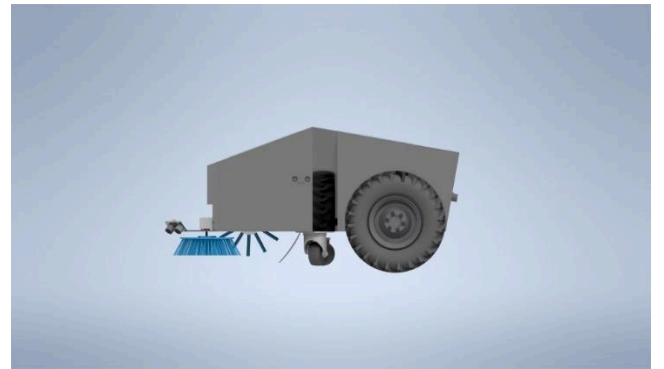
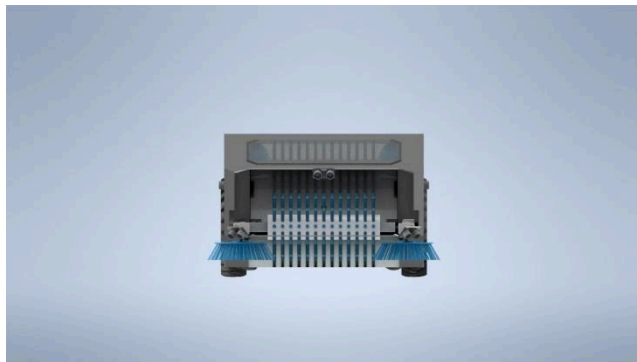
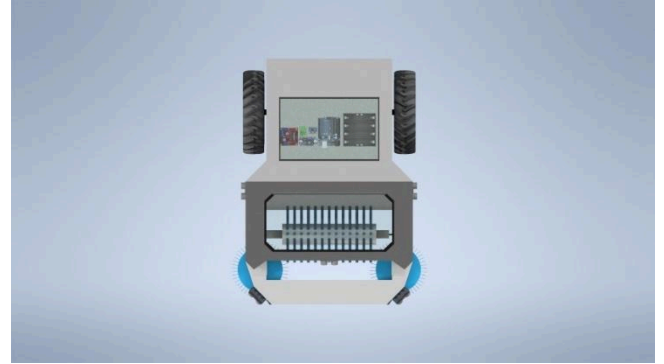
The robot's chassis is constructed using lightweight aluminum sheets and profiles due to their high tensile strength, making them a more suitable choice than acrylic sheets of the same size. The drivetrain features two high-torque 12-volt DC motors that exceed the required torque, enabling the robot to navigate challenging terrains such as tall grass, concrete, sand, and mud effectively. As for the leaf collection mechanism, the design utilizes three brushes powered by the same motors as the drivetrain, ensuring efficient leaf collection on various surfaces. These brushes are made from durable PLA plastic, providing longevity and resistance to wear and tear, minimizing maintenance and extending the robot's operational lifespan. Regarding the electronics, the system employs a configuration of five ultrasonic sensors strategically placed around the vehicle for obstacle detection. Additionally, a compass sensor and an accelerometer are employed to ensure the vehicle's stability. The entire array of electronic components is centrally managed and processed via the Arduino Mega microcontroller.





1.2 Drawing

Initial sketch



2. DESIGN

2.1 Components and Parts

Mechanical Design

Components	Quantity
10x20 V-Slot Aluminum Profile	2000mm
0.8mm 1100 Aluminum Sheet	600x1200mm
10mm-M5 Screws	20x
15mm-M4 Screws	10x
15mm-M3 Screws	20x
45mm-M2 Screws	12x
U-Bolts	4x
Threaded Rods	2x
Various Custom 3D Printed Parts	42x
Custom-Made Brushes	3x

Electrical Design

Components	Quantity
Arduino Mega 2560	1x
L298N Motor Driver	1x
GB37-En448 Motors	5x
21700 4500mAh 20A Li-ion batteries	6x
GY-521 MPU6050 Accelerometer	1x
GY-271 QMC5883L Compass	1x
HC-SR04 Ultrasonic	5x
Jumper Wires	14x
12 Gauge Wires	500mm
16 Gauge Wires	1500mm

2. 2 Sensors

In the sensor selection, the following sensors were chosen:

1. HC-SR04 Ultrasonic Sensor: This sensor was utilized for obstacle detection and avoidance due to its affordability, accessibility, and overall reliability, making it well-suited for the task.
2. GY-521 MPU6050 Accelerometer: Initially intended for crash detection, it was repurposed for angle measurement, adding versatility to its functionality.
3. GY-271 QMC5883L Compass: Used alongside the accelerometer, it aided in detecting the current orientation angle, enhancing the robot's dimensional awareness and navigation capabilities.

The inclusion of two angle-measuring sensors facilitated the implementation of the Kalman Filter, which effectively mitigated unnecessary noise and fluctuations, ensuring the robot's operational efficiency.

2.3 Kinematics

Kinematics Model

Forward kinematics provides an estimate of the robot's position given by its geometry and the speed of its wheels. It can be determined by the Instantaneous Center of Rotation.

$$\varepsilon = [x \ y \ \theta] = f(l, r, \theta, v1, v2)$$

θ = *robot angle*

V = Forward velocity moving in a straight line

$V1$ = Linear Velocity of the right wheel

$V2$ = Linear Velocity of the left wheel

ω = *Angular velocity*

l = Distance from robot center to wheel

R = Radius of ICR (Instantaneous Center of Rotation)

For non-linear displacement ($v1, v2$), assume that the robot is following the ICR with radius R at angular rate ω at each instance of time. ICR changes over time as a function of the individual wheel velocities.

The angular velocity (ω) of the robot during a turn is determined by the difference in speed between its two wheels ($v1 - v2$) divided by twice the distance between the wheels ($2 * l$).

$$\omega = \frac{v1 - v2}{2l}$$

The radius of ICR is determined by how fast the object is moving in a straight line (V) and how quickly it's turning (w).

$$R = \frac{V}{\omega} = \frac{l(V1+V2)}{(V1-V2)}$$

The position can be determined as a function of the velocity and position.

$$p(t + \Delta) = F(v1, v2)p(t)$$

At time $(t + \Delta)$, the robot poses with respect to the ICR is

$$p(t + \Delta) = R(\omega\Delta)^{-1} p(t) + ICR(t)$$

$$p(t + \Delta) = \begin{bmatrix} x(t + \Delta) \\ y(t + \Delta) \\ \theta(t + \Delta) \end{bmatrix} = \begin{bmatrix} \cos(\omega\Delta) & -\sin(\omega\Delta) & 0 \\ \sin(\omega\Delta) & \cos(\omega\Delta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R \sin \theta_t \\ -R \cos \theta_t \\ \theta_t \end{bmatrix} + \begin{bmatrix} ICR_x \\ ICR_y \\ \omega\Delta \end{bmatrix}$$

Since $ICR(t) = (ICR_x, ICR_y) = (x(t) - R \sin \theta, y(t) + R \cos \theta)$,

$$p(t + \Delta) = [x(t + \Delta), y(t + \Delta), \theta(t + \Delta)]$$

$$x(t + \Delta) = R \cos(\omega\Delta) \sin \theta_t + R \sin(\omega\Delta) \cos \theta_t + (x_t - R \sin \theta_t)$$

$$y(t + \Delta) = R \sin(\omega\Delta) \sin \theta_t - R \cos(\omega\Delta) \cos \theta_t + (y_t + R \cos \theta_t)$$

$$\theta(t + \Delta) = (\theta_t + \omega\Delta)$$

For linear displacement ($v1 = v2 = v$), the robot moves in a straight line so it ignores the ICR.

$$x(t + \Delta t) = x_t + v_t \Delta \cos \theta_t$$

$$y(t + \Delta t) = y_t + v_t \Delta \sin \theta_t$$

$$\theta(t + \Delta t) = \theta_t$$

The length of the robot = 22.5cm

Initial position $(x_0, y_0) = (0 \text{ cm}, 0 \text{ cm})$

Turning angle $\theta = 0$ as it is in a straight position.

The robot moves both wheels at 4cm/s and moves for 50s. Its position will become (200cm, 0cm).

$$x(0+\Delta 50) = 0 + 4 \times 50 \cos 0 = 200 \text{ cm}$$

$$y(t+\Delta 50) = 0 + 4 \times 50 \sin 0 = 0 \text{ cm}$$

$$\theta(t+\Delta 50) = 0 \text{ degree}$$

The length of the robot = 22.5cm

Initial position $(x_0, y_0) = (200 \text{ cm}, 0 \text{ cm})$

Turning angle $\theta = 0$ as it is in a straight position.

Now the robot sets the right wheel to 4cm/s and the left wheel to 3cm/s and moves for 20 more seconds.

$$\omega = \frac{v_1 - v_2}{2l} = \frac{4-3}{2(22.5)} = 0.022 \text{ rad/s}$$

$$R = \frac{22.5(4+3)}{(4-3)} = 157.5 \text{ cm}$$

$$\begin{aligned} x(50+20) &= 157.5 \cos (0.022 \times 20) \sin 0 + 157.5 \sin (0.022 \times 20) \cos 0 + (200 - 157.5 \sin 0) \\ &= 157.5 \sin(25.21 \text{ degree}) + 200 = 334.17 \text{ cm} \end{aligned}$$

$$\begin{aligned} y(50+\Delta 20) &= 157.5 \sin (0.022 \times 20) \sin 0 - 157.5 \cos (0.022 \times 20) \cos 0 + (0 + 157.5 \cos 0) \\ &= -157.5 \cos (25.21) + 157.5 = 15 \text{ cm} \end{aligned}$$

$$\theta(50+\Delta 20) = 0.022 \times 20 = 0.44 \text{ rad} = 25.21 \text{ degree}$$

The final position will become (334.17 cm, 15 cm, 25.21 degree)

2.4 Dynamics Model

Mass of the car = 3.5kg

$$\text{Acceleration from kinematic model} = \frac{\text{Change in Velocity } \Delta V}{\text{Time}} = \frac{4-0}{50} = 0.08 \text{ cm/s}^2 = 8 \times 10^{-4} \text{ m/s}^2$$

Force required to move the car, $F = ma$

$$= 28 \times 10^{-4} \text{ N (2 motors)}$$

Force required from each motor, $F_m = 14 \times 10^{-4} \text{ N}$

Radius of the sprocket = 30mm = 0.03m

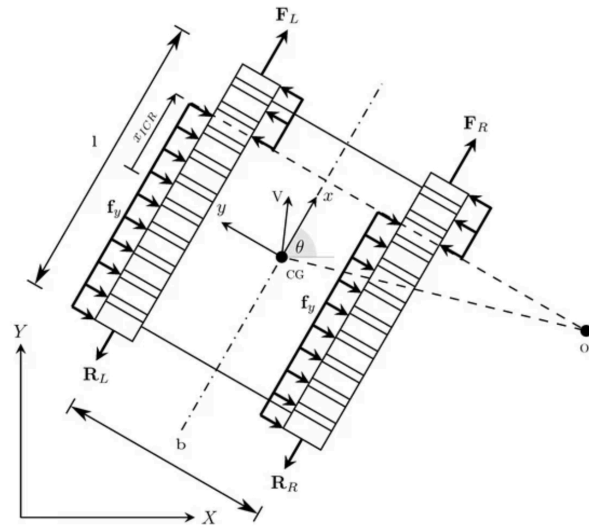
Torque required of each motor, $\tau = F_m \times \text{Radius of the sprocket}$

$$= 14 \times 10^{-4} \times 0.03 = 42 \times 10^{-6} \text{ Nm}$$

Torque of GB37-En448 motor , $\tau : 8.8 \text{ kgcm}$

Torque of the motor (calculation) < Torque of the motor (GB37-En448)

Forces and Moments Acting on the Tracked Vehicle



A moving vehicle is subjected only to track-terrain interaction forces, which can be classified as follows:

Tractive forces F_R and F_L

Longitudinal resistance forces R_R and R_L

Lateral forces $F_{y,R}$ and $F_{y,L}$

Moment of turning resistance M_r induced by the resistance forces

Longitudinal resistance forces R_R and R_L

$$r_1 = v_1 mg / 2$$

v_1 = coefficient of Longitudinal resistance

g = gravitational acceleration

The direction of the longitudinal friction forces

$$\begin{aligned} G(F, f, \bar{x}) &= [-f \operatorname{sign}(\bar{x}), \bar{x} \neq 0] \\ &= [0, \bar{x} = 0, |F| \leq f] \\ &= [-f \operatorname{sign}(F), \bar{x} = 0, |F| > f] \end{aligned}$$

Longitudinal resistance force can be defined as :

$$R_{l,i} = G(F_i, r_1, v_{t,i}), i = \{R, L\}$$

Lateral friction distribution and force can be obtained :

$$\begin{aligned} F_y &= \frac{v_t m g}{l}, F_y = -2 \operatorname{sign}(\bar{y}) f_y X_{ICR} \\ v_t &= \text{coefficient of lateral resistance} \end{aligned}$$

Turning Moment and Moment of turning resistance

$$M = (F_L - R_{l,L})b/2 - (F_R - R_{l,R})b/2 = (F_L - F_R)b/2$$

Resistance moment can be obtained by integrating over the track length of the distribution:

$$m_r = 2f_y \left(\frac{l^2}{4} - x_{ICR}^2 \right)$$

Direction of moment of turning resistance

$$M_r = G(M, m_r, \bar{\theta})$$

Drive Model

The truck is actuated with two DC motors that drive the sprockets through gear.

The relationship between the motor torque τ and the rotor current i_a is linear.

$$\tau_m = k_m i_a$$

k_m is the motor torque constant

Voltage and current in the motor can be approximated with a differential equation.

$$u_a = L_a \frac{d}{dt} i_a + R_a i_a + k_e \omega_m$$

L_a, R_a are the inductance and resistance of the motor

k_e is electromotive force coefficient

ω_m is the rotational speed of the sprocket

$$\tau = n k_m i_a$$

Motor control input, $u_a = L_a \frac{d}{dt} i_a + R_a i_a + k_e \omega_m$

$$\tau = [\tau_L \ \tau_R]^T, u_a = [u_L \ u_R]^T$$

$$i_a = [i_L \ i_R]^T$$

$$\omega = [\omega_L \ \omega_R]^T$$

2.5 Control System

The system is equipped with five inputs, including five ultrasonic sensors, a compass module and an accelerometer sensor. The fundamental principle of this system revolves around utilizing data from the five ultrasonic sensors to perform Kalman filtering. The software module reads distance measurements from the ultrasonic sensors and employs a predictive algorithm based on prior state estimations and the system's dynamic model. The Kalman filter is instrumental in forecasting the present state of the system, along with an associated estimation of state uncertainty.

Subsequently, as new sensor measurements become available, the Kalman filter engages in a comparison between the predicted state and the actual measurement, culminating in the computation of the Kalman gain. This gain signifies the relative significance of the prediction versus the measurement and is subsequently utilized to fuse the prediction and measurement data, leading to an updated state estimate characterized by reduced uncertainty. This process iterates continuously as new measurements are obtained, refining the system's state estimate over time.

3. IMPLEMENTATION

3.1 Hardware

The team engaged in thorough discussions and collaborative idea generation to devise the most suitable hardware configuration for our robot. This involved ensuring that the robot's structure met the necessary criteria, alongside designing the electronic components to maximize safety and efficiency.

Materials

In the construction of the robot hardware, aluminum beams were employed as the primary structural elements, providing essential support for the overall framework. Regarding the external shell, utilizing Aluminum sheets was driven by considerations of cost-effectiveness, lightweight properties, and durability, in addition to its aesthetic compatibility with the intended design vision, distinguishing it from alternative materials such as acrylics and cardboard.

The wheels were exclusively crafted using 3D printing technology, as suitable pre-existing tracks meeting our specific requirements were not commercially accessible. Moreover, a small amount of 3D printed material was utilized to facilitate the convenient assembly of the robot's black panel.

Shell Design

As previously mentioned, the shell design was influenced by Tesla's Cybertruck, primarily due to its low center of mass, which facilitated the integration of various components onto the robot, including electronic elements and sensors. The curved top was also incorporated to prevent dents, thereby preserving the robot's appearance and protecting intricate components from impacts, ultimately prolonging its lifespan.

Tracks

Originally, the design concept featured the inclusion of two wheels on each side of the robot. However, subsequent reconsideration led to a shift towards tracks, primarily driven by their superior capability to traverse soft terrains, a critical requirement for the project.

Research findings revealed the effectiveness of triangular tracks in maneuvering through muddy or soft work environments, highlighting their inherent flexibility. Consequently, the decision was made to utilize tracks with this particular geometric configuration for the project.



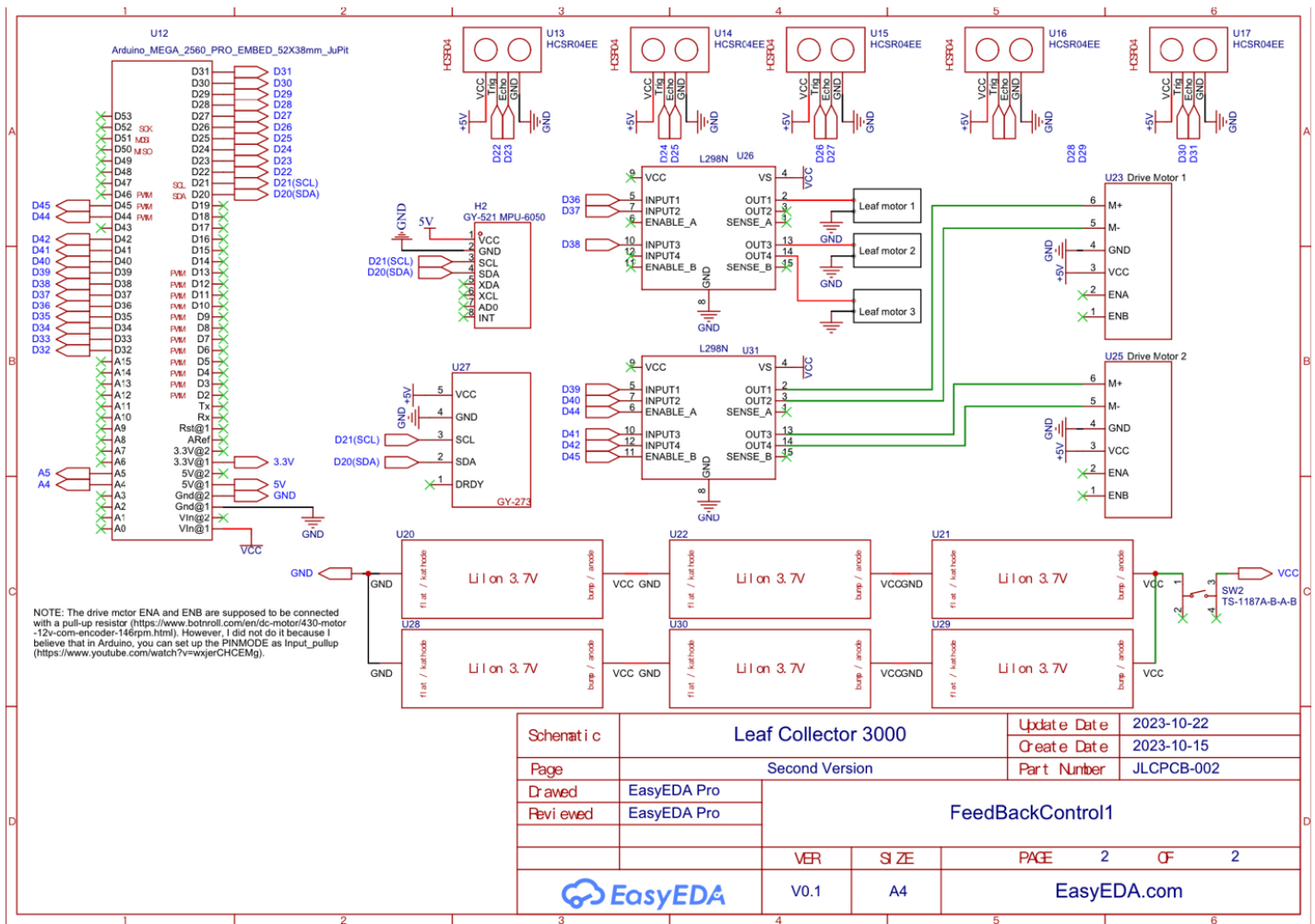
(Triangle track's performance on soft surfaces)



(Physical robot's track flexibility test)

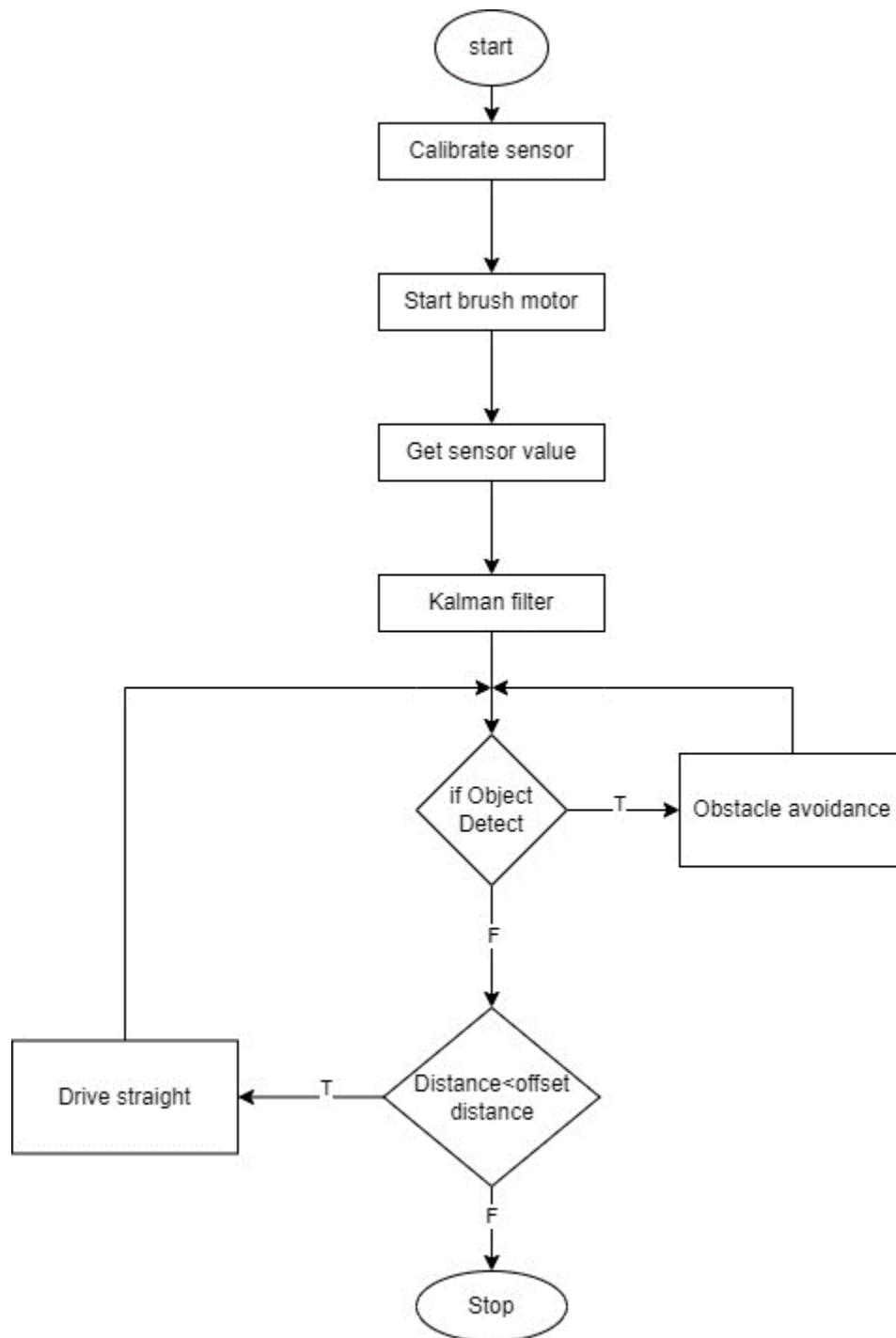
Electric circuit

The challenges encountered in the electric circuitry aspect primarily revolved around insufficient voltage to power the motors. To address this issue, we utilized six 3.7V batteries, configuring them with three batteries in series per row to increase the voltage. Subsequently, these rows were interconnected in parallel to boost the current. This specific battery arrangement was chosen as it enabled us to compensate for the disparity between the motor's requirement of 12V and the available resources. Charging each battery to its maximum capacity resulted in an individual voltage increase to 4.2V, ultimately providing a total of 12.6V per row. Consequently, the system successfully fulfilled the required voltage and current criteria.



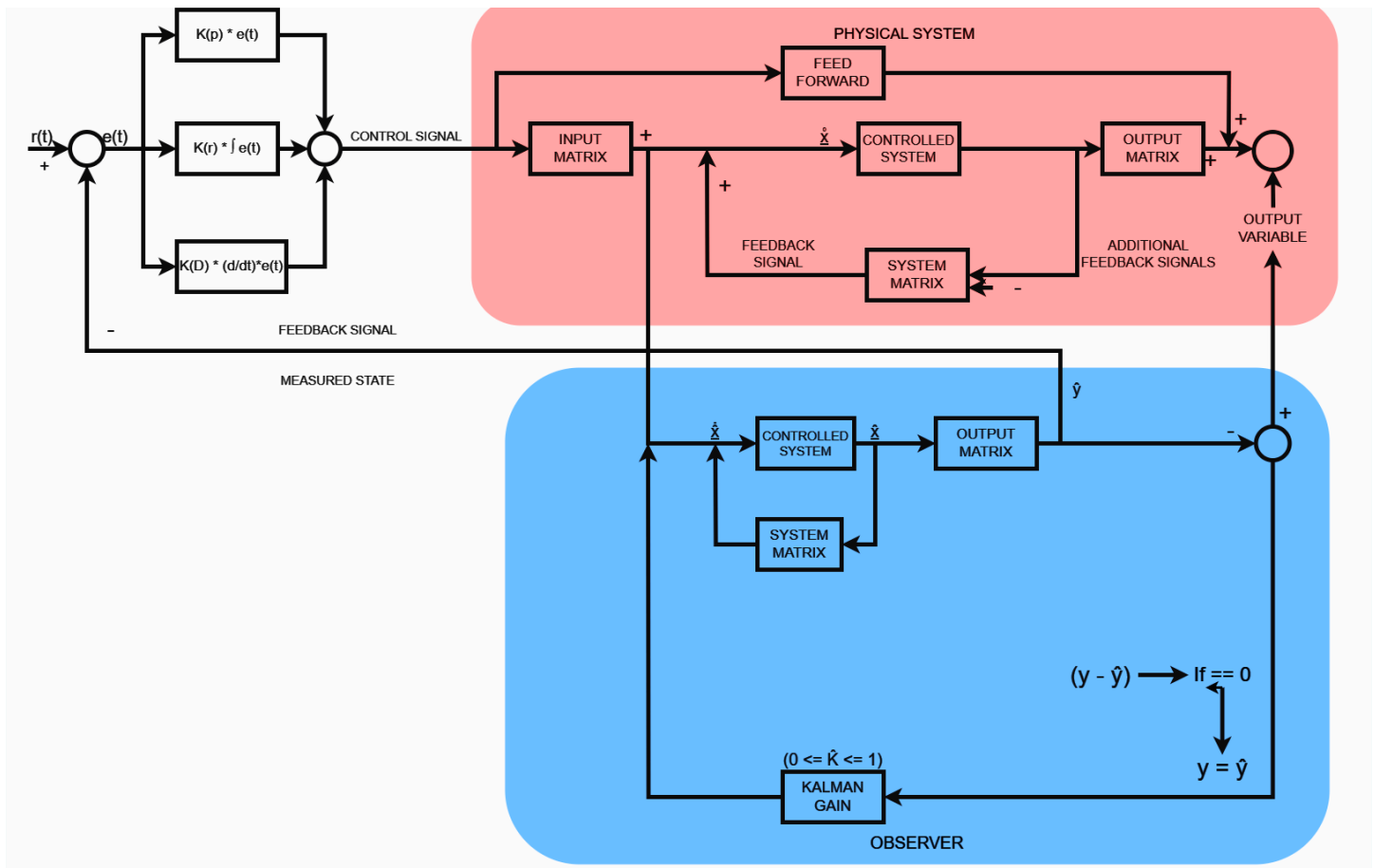
Schematic of the robot

3.2 Software



(Program Flowchart)

3.3 Block Diagram



4. TEST RESULT

4.1 Kalman Filter

The Kalman Filter is an arithmetic process used to reduce noise and stabilize fluctuations from given inputs by efficiently predicting the current state of the system and the uncertainty of the predictions to then correct the values thus eliminating the unnecessary noises accurately, we applied this to both of our concurrently working angle sensors, the accelerometer and the compass sensor to get the best and utmost accurate values.

Equations for Kalman Filter

1. Predict the current state of the system:

$$S(k) = F \cdot S(k - 1) + G \cdot U(k)$$

S = state vector (Anglekalman) F = state transition matrix (1) G = control matrix U = rate of input variable

2. Calculate uncertainty of the Prediction

$$P(k) = F \cdot P(k - 1) \cdot F^T + Q$$

P = prediction uncertainty vector (Uncertaintyangle) Q = process uncertainty ($T^2, \cdot 4^2$)

3. Calculate the Kalman Gain from the uncertainties on the predictions and measurements

$$L(k) = H \cdot P(k - 1) \cdot F^T + R$$

L = Intermediate matrix K = Kalman gain H = Observation $T^2, \cdot 3^2$)

4. Update the predicted state of the system with the measurement Of the state through Kalman Gain

$$S(k) = S(k) + K \cdot (M(k) - H \cdot S(k))$$

M = measurement vector (angle from MPU)

5. Update the uncertainty of predicted area

$$P(k) = (I - K \cdot F) \cdot P(k)$$

$I = \text{unit matrix } (=I)$

```
void kalman_1d(float KalmanState, float KalmanUncertainty, float KalmanInput, float KalmanMeasurement) {
    unsigned long Tnew = millis();
    float dt = (Tnew - preInterval) * 1e-3;
    preInterval = Tnew;
    KalmanState=KalmanState+dt*KalmanInput;
    KalmanUncertainty=KalmanUncertainty + dt * dt * 4 * 4;
    float KalmanGain=KalmanUncertainty * 1/(1*KalmanUncertainty + 10 * 4); //4*4 is also cool
    KalmanState=KalmanState+KalmanGain * (KalmanMeasurement-KalmanState);
    KalmanUncertainty=(1-KalmanGain) * KalmanUncertainty;
    Kalman1DOutput[0]=KalmanState;
    Kalman1DOutput[1]=KalmanUncertainty;
}
```

(Code implementation)

The input for the “KalmanInput” variable in this case would be the MPU and a compass for the “KalmanMeasurement” variable.



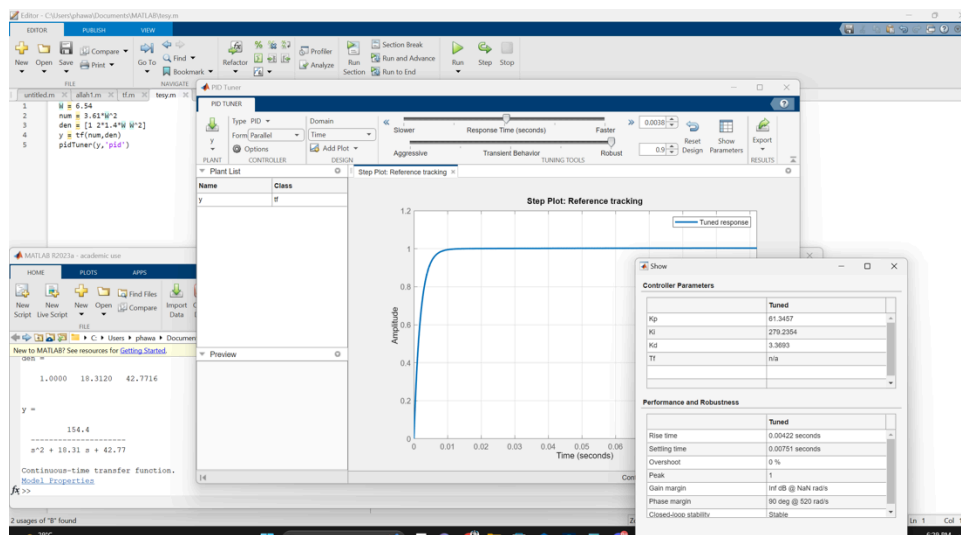
(Serial plotter implementation)

The blue and orange lines show the measured angle of the Compass and MPU sensors in real-time respectively, the green line shows the filtered angle, calculated using the values from the aforementioned sensors.

4.2 PID

The process of PID tuning is to essentially reduce the response time of the system to the lowest we possibly can.

For our PID tuning we took the values from the Arduino board such as the rise time from the serial plotter, PWM counts, The standard degree with rise time to find omega then use that to find the transfer function in MatLab to then tune so that we have the best Rise time and Setting time as possible:



(MatLab Tuning)

The Kp, Ki, and Kd values are then implemented into the code to ensure the best response time for our code.

```

int x_value;
int y_value;
int z_value;
int azimuth; // 0° - 359°
byte bearing; // 0 - 15 (N, NNE, NE, ENE, E, ...)
char direction[strlen("NNE") + 1];
int firstAngle, currentAngle;
uint32_t LoopTimer;
float KalmanAngleRoll=0, KalmanUncertaintyAngleRoll=2*2;
float Kalman1DOutput[]={0,0};
float Roll;
long preInterval;
int P, D, I, previousError, PIDvalue, error;
int lsp, rsp;
int lfspeed = 150;

float Kp = 50;
float Kd = 0.5;
float Ki = 0.1;

```

(Implementation onto code)

4.3 Stability Analysis

For the stability analysis we basically observed the sign changes in the transfer function and we noticed that it is completely stable due to the fact that there are no sign changes.

$$y(s) = \frac{154.5}{s^2 + 18.375s + 42.77}$$

s^2	1	42.77	0
s^1	18.37	0	-
s^0	$\frac{- 1 \ 42.77 \ 18.37 \ 0 }{18.37}$	0	-

No sign changes = stable

5. CONCLUSION

5.1 Concluding Remarks

In summation, the project offered us a slew of hurdles and challenges that, in retrospect, considerably aided our understanding of key class concepts. This hands-on experience acted as a bridge between theoretical understanding and practical application, allowing us to effectively implement design principles and calculative approaches in real-world circumstances to maximize our robot's performance. These difficulties aided our academic and personal development by providing us with essential skills and insights that will certainly be useful in our future endeavors.

5.2 Suggestion

Upon reevaluation, it is advisable to consider several enhancements for prospective iterations as followed;

- The magnetometer compass sensor should be substituted with a more robust and precise alternative, such as a smartphone that can establish a wireless connection with the controller board. This modification would mitigate the magnetic interference emanating from the robot's chassis and circuitry, thereby enhancing navigation accuracy.
- The incorporation of larger track wheels into the design is recommended to bolster the robot's capability to navigate particularly challenging terrains effectively. An illustrative example is the robot's struggle in traversing areas with tall grass.
- The integration of a removable leaf collector box within the robot's chassis would facilitate the uncomplicated disposal of collected leaves and reduce the necessity for manual intervention.

6. TESTING VIDEO

Link: https://drive.google.com/file/d/13585rS_QGzy9cnES0mCGRXDxx3RUGiNv/view